COMPASS Synoptic CB: Sulfide

Sept 2026 Maryland Sample Reruns

2026-10-13

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
  #And Will Write a Rerun File

###things that need to be changed
sample_month = "Sept"
sample_year = "2026"
Sample_Month_Year = "202609"
Run_Date = as.Date(c("10/07/2026"), format = "%m/%d/%Y")
Run_by = "Zoe Read"#Instrument user
Script_run_by ="Zoe Read"#Code user

Run_notes=""#any notes from run

#Plate Data
folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE,pattern = "Plate.*20261007\\.csv")
(Raw_plates)
```

```
## [1] "Raw Data/COMPASS_H2S_Sept2026_Plate1_20261007.csv"
## [2] "Raw Data/COMPASS_H2S_Sept2026_Plate2_20261007.csv"
```

```
# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Raw Data/Sulfide_STD_QAQC.csv"

# Metadata path
Raw_Metadata <- "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2026.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/COMPASS_SynopticCB_H2S_202607_MD_reruns_processed.csv"
rerun_path <- "Processed Data/COMPASS_SynopticCB_H2S_202607_MD_reruns.csv"
```

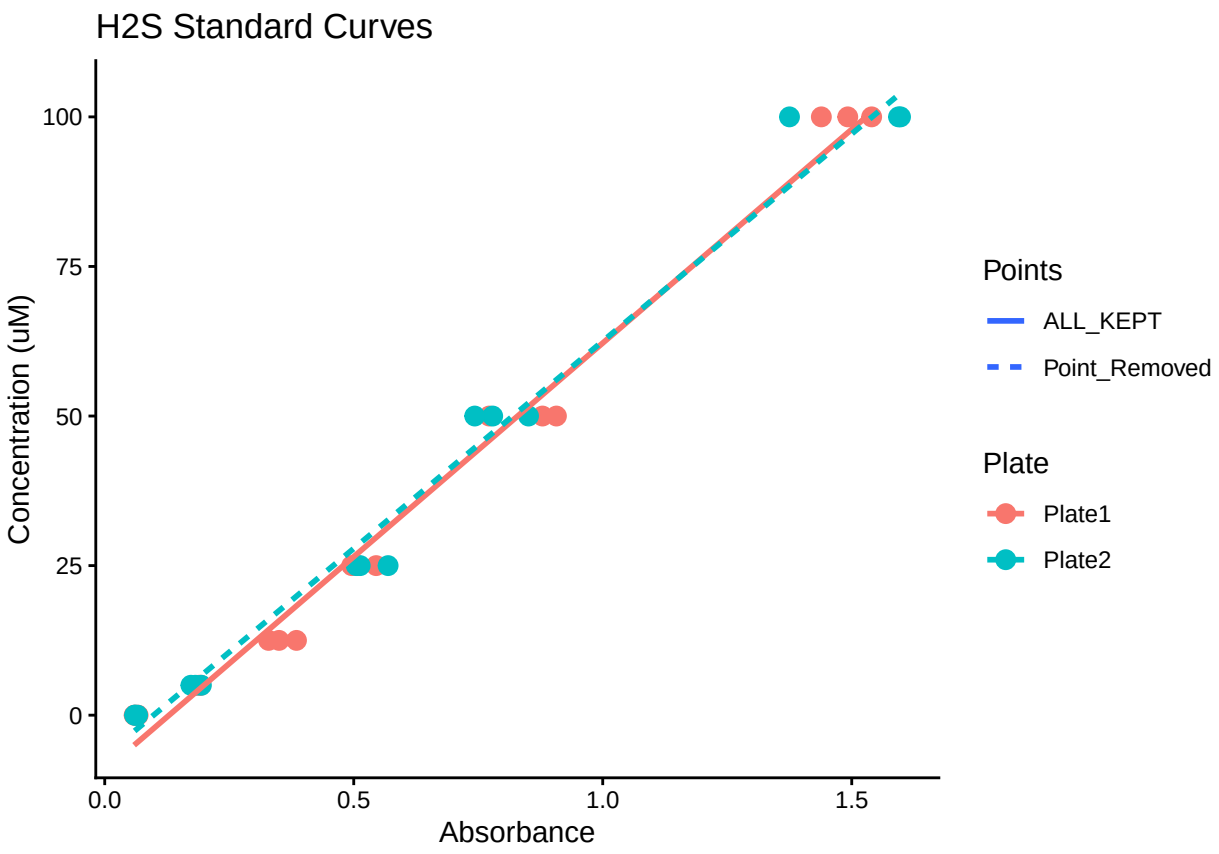
##Read in data

##Remove High CV Stds

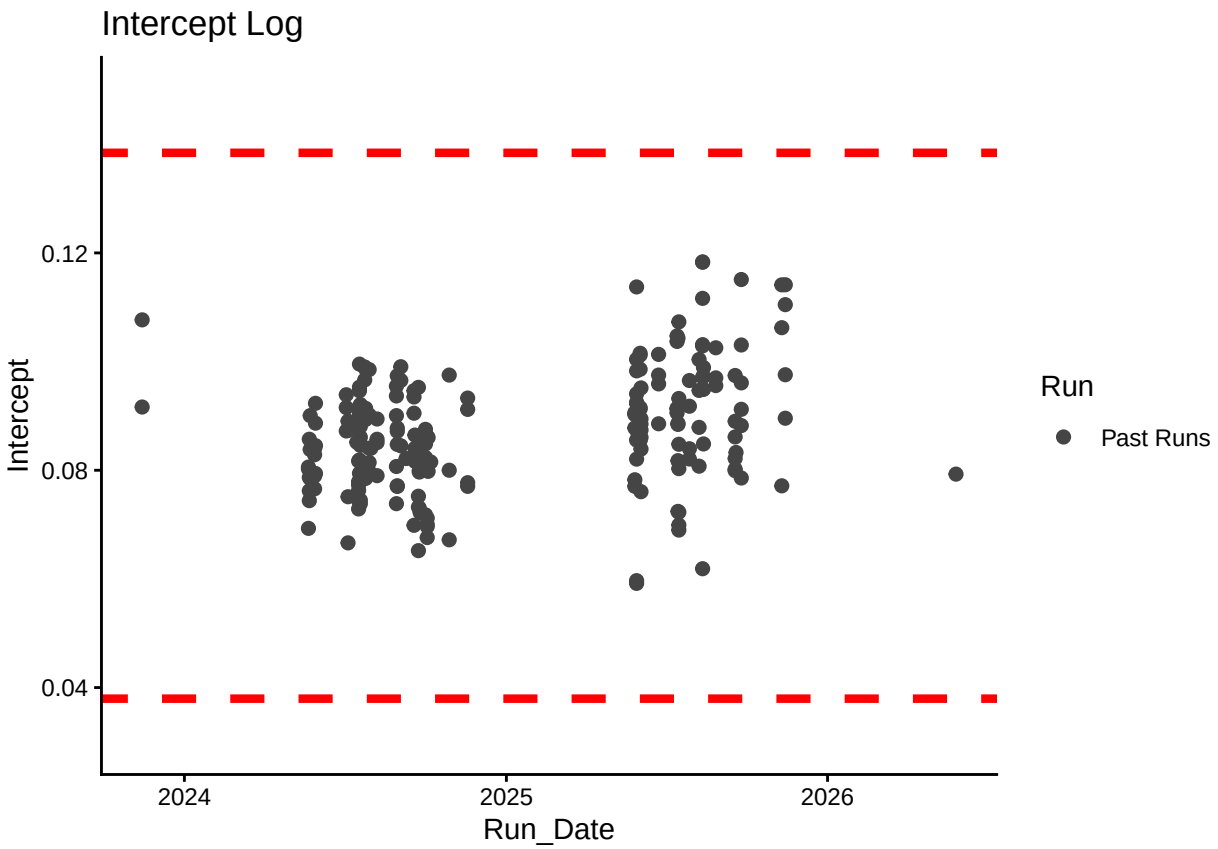
##Check R2 & Remove Standards if Necessary

```
## Joining with `by = join_by(Plate, IDs)`
```

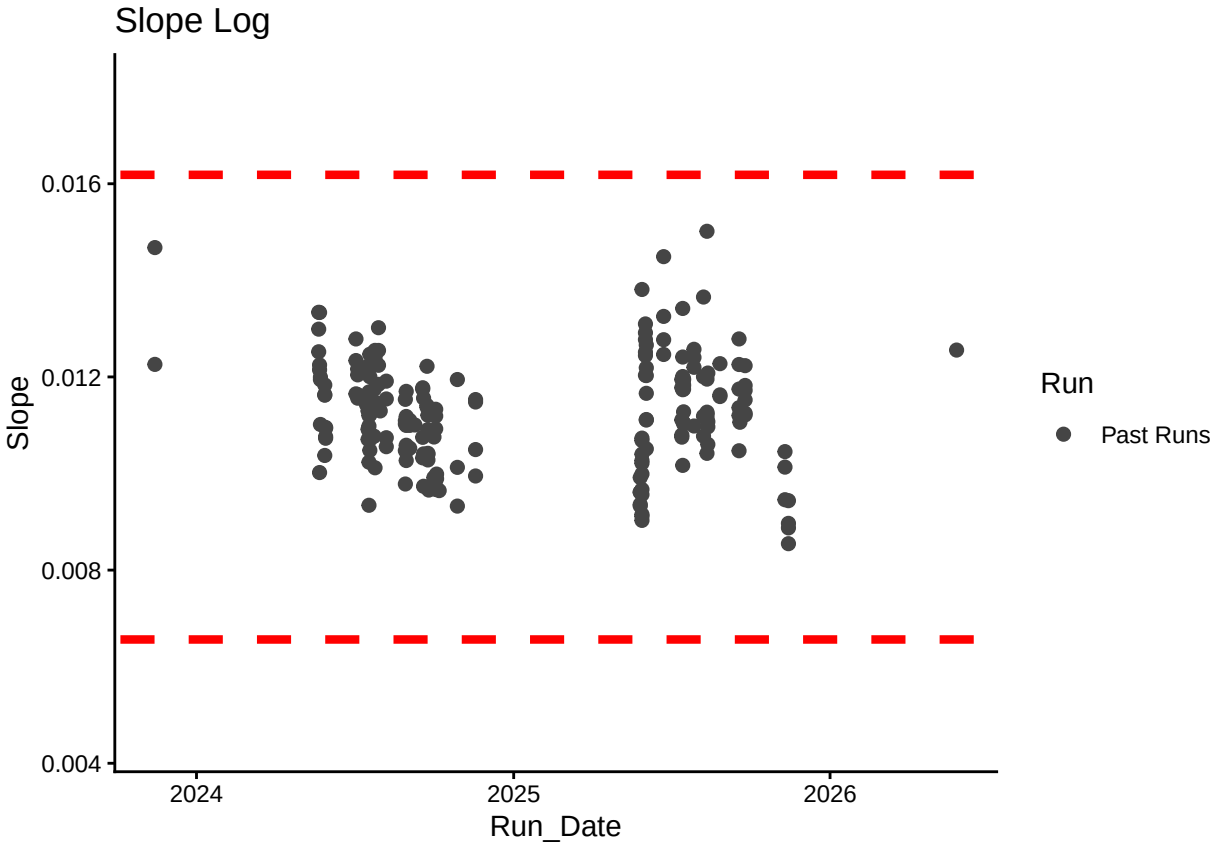
Plot Standard Curves




```
## Warning: Removed 17 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
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## (`geom_point()`).
```

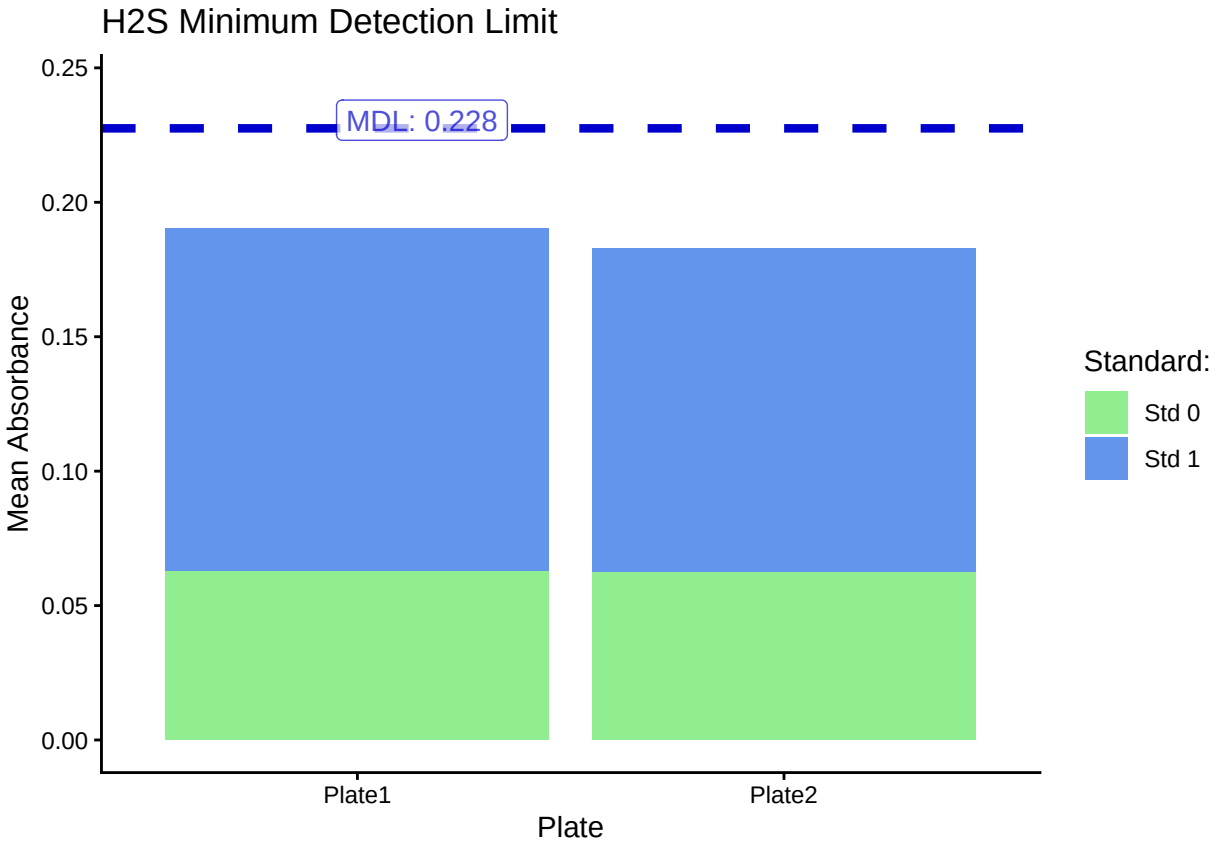


Return Flags For Standard Curve

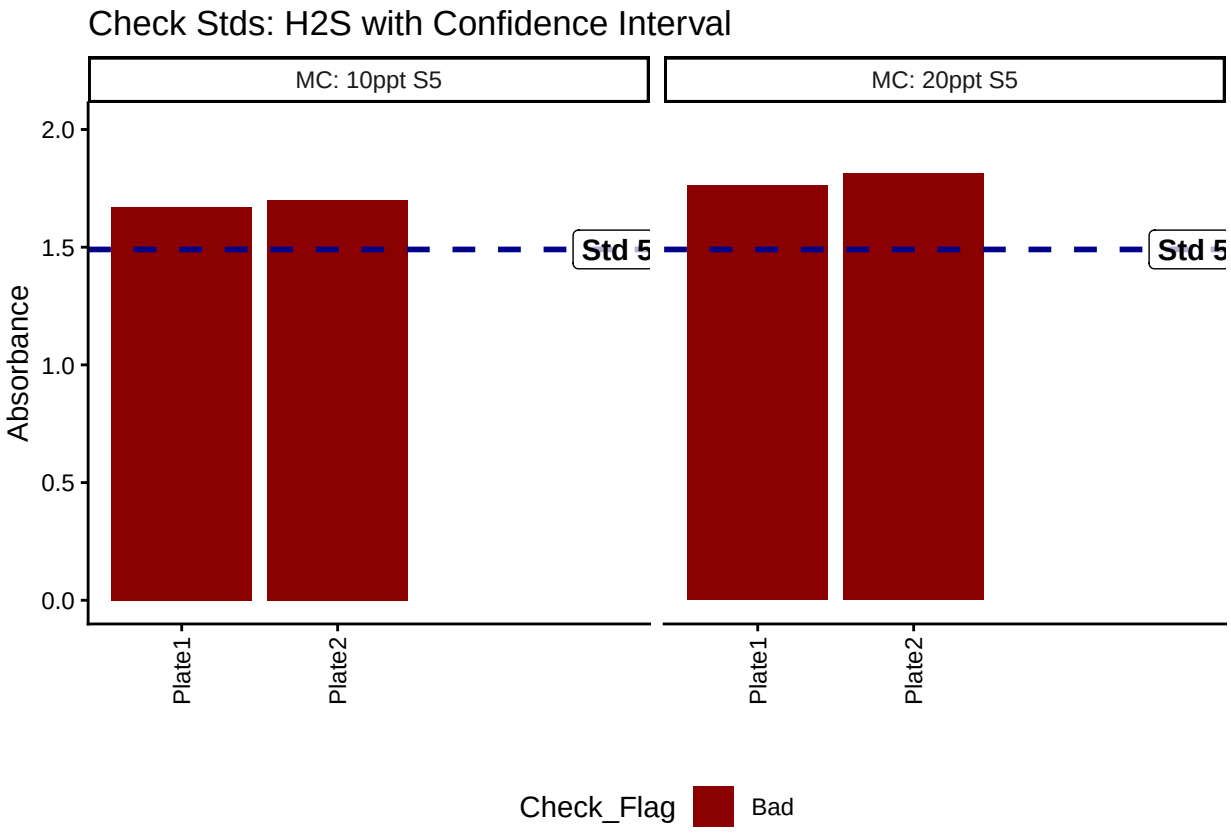
Plate	R2_Flag	CV_Flag
Plate1	R2_Good	CV_Good
Plate2	R2_Bad	CV_Good

Plate	Curve_RMVD
Plate1	KEPT
Plate2	REMOVED

Method Detection Limit



Compare Check Standards to Standards



Display Any Check Flags

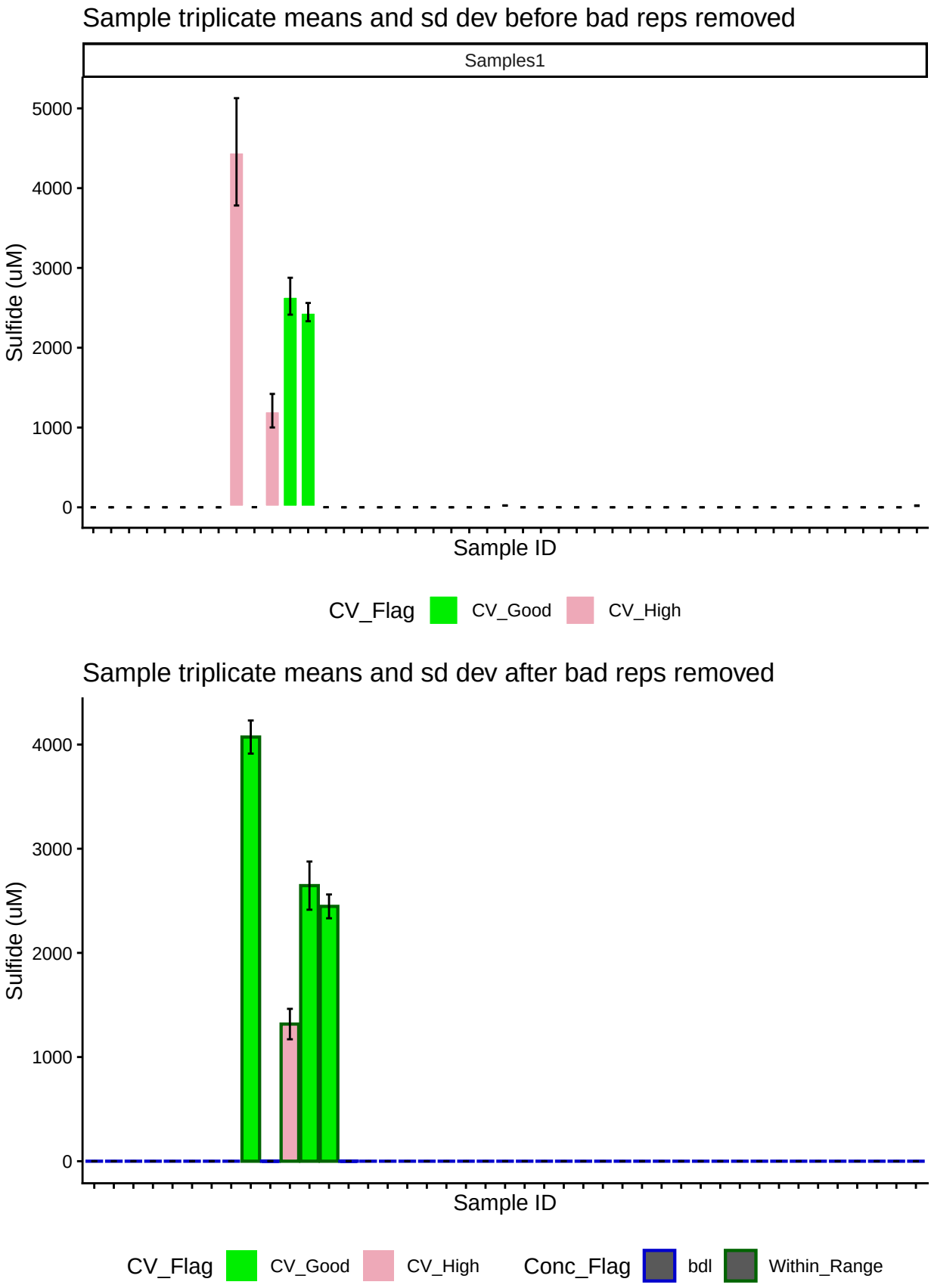
Plate	CHKs_Flag
Plate1	MC: 10ppt S5 Chk_Bad
Plate1	MC: 20ppt S5 Chk_Bad
Plate2	MC: 10ppt S5 Chk_Bad
Plate2	MC: 20ppt S5 Chk_Bad

Plate	CHK_CV_Flags

##Calculate Sulfide Concentrations & Add Concentration Flags

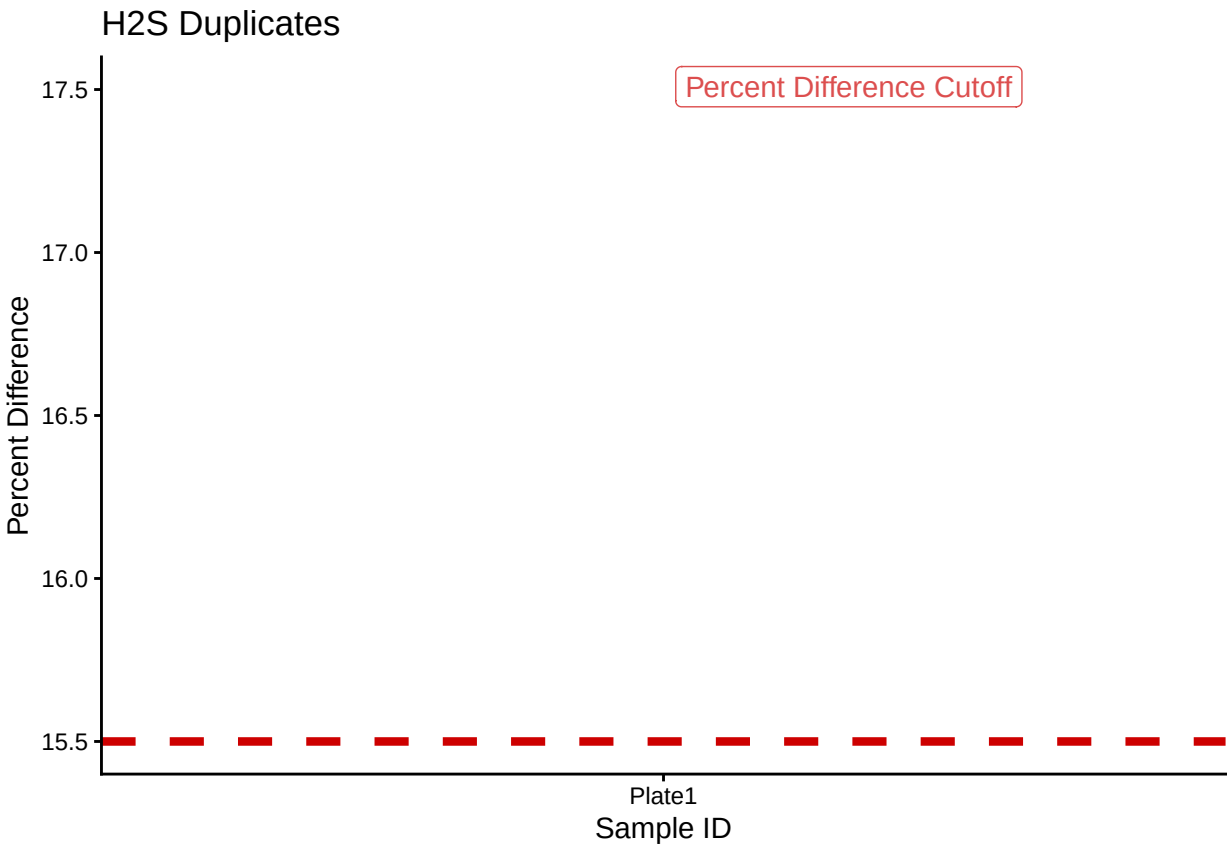
##Remove High CV Samples and Average

Plot Samples



Duplicate QAQC

```
## Warning: No shared levels found between ‘names(values)‘ of the manual scale and the
## data’s fill values.
```

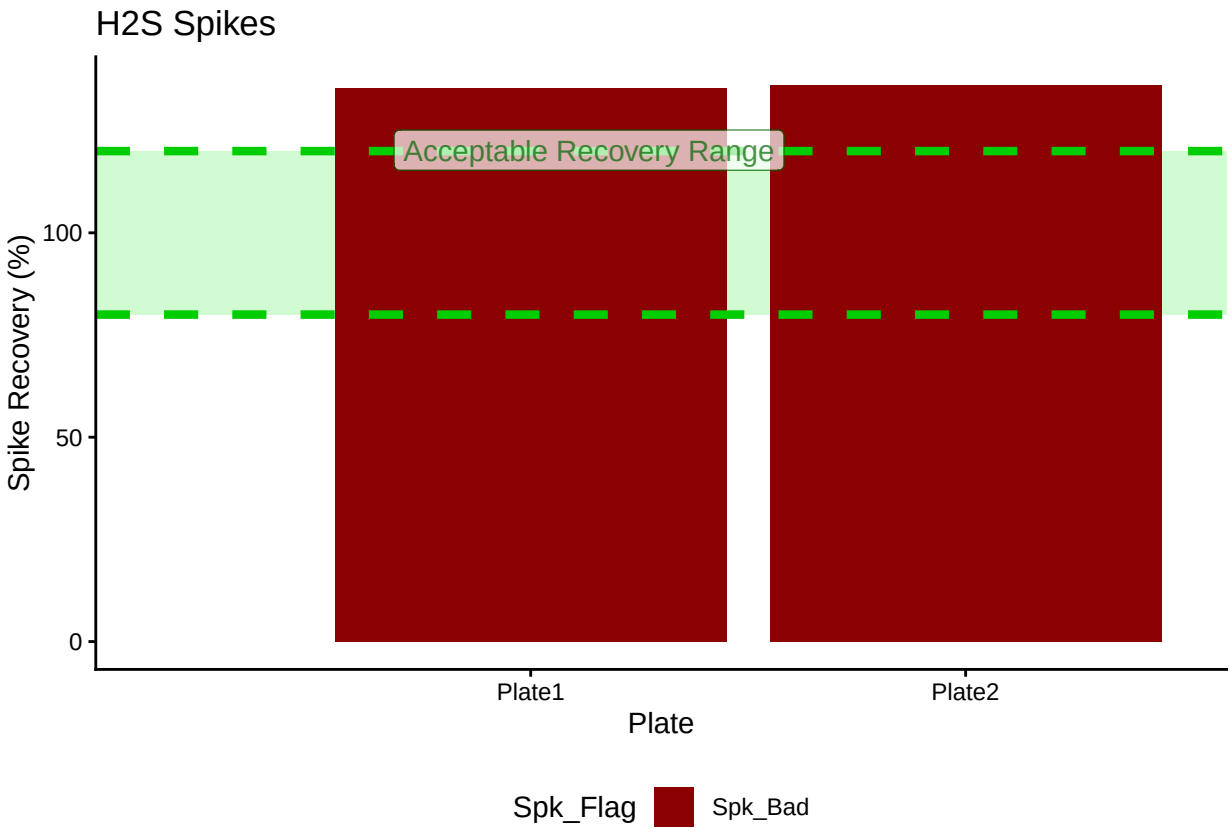


Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate1	Dup_Good
Plate2	Dup_Good

Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate1	Spk_Bad
Plate2	Spk_Bad

```
## [1] "<60% of Spikes Pass, Rerun"
```

Check for reruns and replace

##Add Flags

##Remove Synoptic samples and rename TEMPEST samples

```
df_all_clean <- df_all_clean1 %>%
  filter(!str_detect(Sample_ID, "SWH|GCW|GWI|MSM")) %>%
  filter(!str_detect(Sample_ID, "FATE"))  %>%
  filter(!str_detect(Sample_ID, "dup"))  %>%
  filter(!str_detect(Sample_ID, "spk"))

##TMP_C_C6_50_20260618 --> TMP_FW_PW_C6_50cm_SO4_20260618
Sample_data_IDs <- df_all_clean %>%
  mutate(Plot = stringr::str_extract(Sample_ID, 'SW|FW|_C_|ESTUARY'),
```

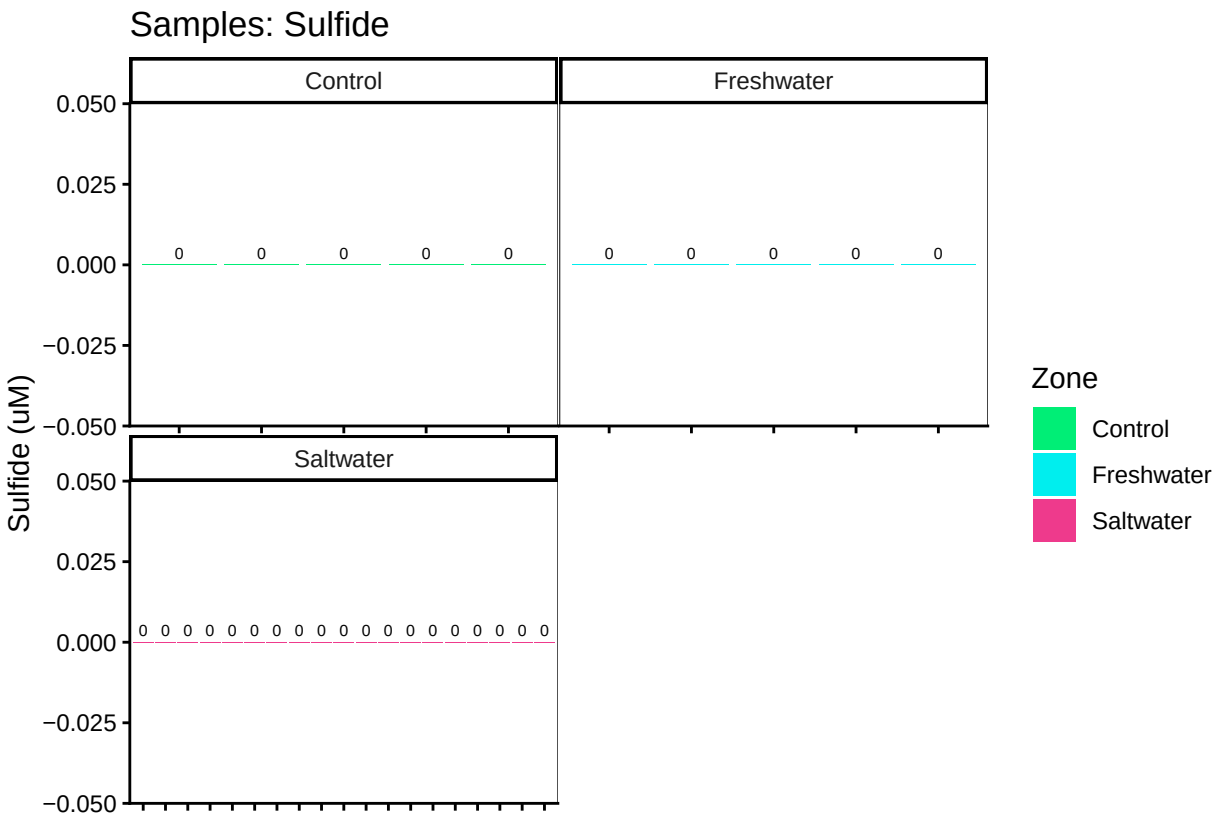
```
        Plot = gsub("_", "", Plot),
        Grid = stringr::str_extract(Sample_ID, "B4|C3|C6|D5|E3|F4|F6|H3|H6|I5") ,
        Sample_Type = stringr::str_extract(Sample_ID, "SOURCE|RAINWATER"),
        Sample_Type = replace_na(Sample_Type, "PW"),
        Date = stringr::str_extract(Sample_ID, "[0-9]{8}"),
        Depth = stringr::str_extract(Sample_ID, "_15_|_30_|_50_"),
        Depth = gsub("_", "", Depth),
        Depth = paste0(Depth, "cm")) %>%
  mutate(Sample_ID = paste("TMP", Plot, "PW", Grid, Depth, "SO4", Date, sep = "_"))
```

##Pull in active porewater inventory

##Check samples against metadata

```
## ***All sample IDs are present in metadata.***
```

Visualize Data by Plot



##Export Data

END